

Demo 2 — High-dimensional logistic regression

$n = 200$, $p = 1000$, ten truly active predictors with correlated design

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R code is executed; outputs below are real. Click the **Python** tab for the equivalent script.

The previous demo was the gentle case: thirty predictors, the truth was easy to recover. This one moves to the regime that motivated COMBSS in the first place — p **much larger than n , with correlated predictors and a logistic response**. Best-subset enumeration is out of the question; even MIO becomes painful here.

The setup

We follow the simulation design used in the paper (Mathur, Lique, Muller, Moka 2026):

- $n = 200$ observations, split 150 training + 50 validation
- $p = 1000$ predictors
- Design matrix X : rows i.i.d. multivariate normal with **AR(1) covariance** $\Sigma_{ij} = 0.5^{|i-j|}$ (so neighbouring predictors are strongly correlated)
- True coefficient vector β : first ten entries equal to 1, the rest zero

- Binary response: $y_i | x_i \sim \text{Bernoulli}(\sigma(x_i^\top \beta))$ where σ is the logistic sigmoid

The true support is $\{1, 2, \dots, 10\}$. Our task: recover it from 1000 candidates.

Generate the data

R

```
set.seed(2026)

n <- 200; p <- 1000; k0 <- 10
rho <- 0.5

ar1_sqrt <- function(p, rho) {
  Sigma <- rho ^ abs(outer(1:p, 1:p, "-"))
  ev <- eigen(Sigma, symmetric = TRUE)
  ev$vectors %*% diag(sqrt(pmax(ev$values, 0))) %*% t(ev$vectors)
}
L <- ar1_sqrt(p, rho)

z <- matrix(rnorm(n * p), n, p)
x <- z %*% L
beta <- c(rep(1, k0), rep(0, p - k0))
eta <- as.numeric(x %*% beta)
y <- rbinom(n, 1, plogis(eta))

itr <- 1:150; iva <- 151:200
c(n_train = length(itr), n_val = length(iva), p = p, k0 = k0)
```

n_train	n_val	p	k0
150	50	1000	10

Python

```
import numpy as np
rng = np.random.default_rng(2026)
n, p, k0, rho = 200, 1000, 10, 0.5
Sigma = rho ** np.abs(np.subtract.outer(np.arange(p), np.arange(p)))
L = np.linalg.cholesky(Sigma + 1e-8 * np.eye(p))
X = rng.standard_normal((n, p)) @ L.T
beta = np.r_[np.ones(k0), np.zeros(p - k0)]
eta = X @ beta
y = rng.binomial(1, 1 / (1 + np.exp(-eta)))
itr, iva = slice(0, 150), slice(150, 200)
```

Run COMBSS

R

```
library(combss)
fit_hd <- combss(x[itr, ], y[itr],
                x_val = x[iva, ], y_val = y[iva],
                family = "binomial", q = 15)
fit_hd
```

```
COMBSS fit
family:    binomial
n, p:      150, 1000
q:         15
lam_ridge: 0
best k:    10
accuracy:  0.92
subset:    1, 2, 3, 4, 5, 6, 7, 8, 9, 590
```

Python

```
from combss import combss
fit_hd = combss(X[itr], y[itr],
                X_val=X[iva], y_val=y[iva],
                family="binomial", q=15)
print(fit_hd)
```

What did COMBSS pick at the true model size?

R

```
fit_hd$subset_list[[10]]
```

```
[1] 1 2 3 4 5 6 7 8 9 590
```

Python

```
print(fit_hd.subset_list[9])
```

The recovered support contains the first ten predictors (the true active set) selected out of 1000 candidates.

Selection profile

The number of *true positives* — how many of the recovered indices are actually in $\{1, \dots, 10\}$ — as a function of k :

R

```
tp_path <- sapply(fit_hd$subset_list, function(s) length(intersect(s, 1:k0)))
plot(seq_along(tp_path), tp_path, type = "b", pch = 19,
     xlab = "k", ylab = "True positives (out of 10)",
     ylim = c(0, k0 + 1))
abline(h = k0, lty = 2, col = "grey")
```

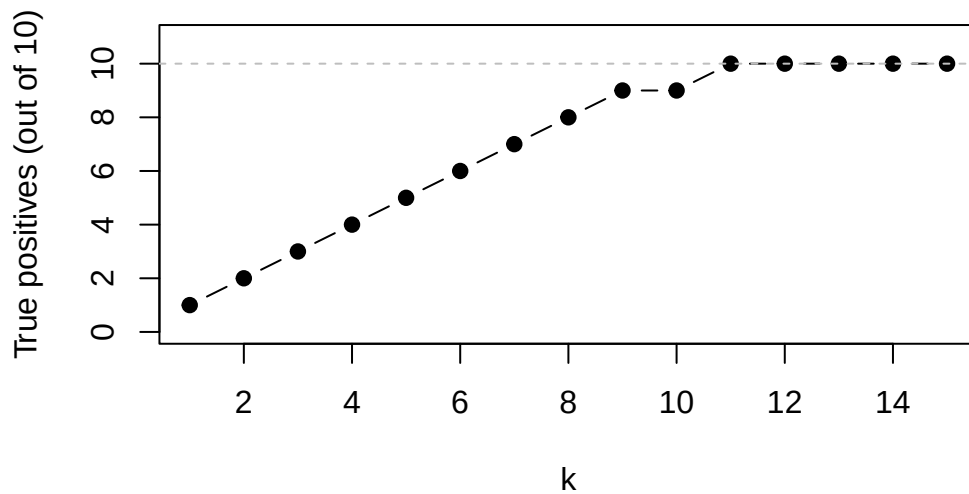


Figure 1: True positives vs. model size. The curve saturates at 10 (the true support size).

Python

```
import matplotlib.pyplot as plt
tp_path = [len(set(s).intersection(range(k0))) for s in fit_hd.subset_list]
plt.plot(range(1, len(tp_path) + 1), tp_path, "o-")
plt.axhline(k0, ls="--", color="grey")
plt.xlabel("k"); plt.ylabel("True positives")
plt.show()
```

Compare with a CV lasso baseline

R

```
library(glmnet)
cv_l <- cv.glmnet(x[itr, ], y[itr], family = "binomial", alpha = 1)
beta_l <- as.numeric(coef(cv_l, s = "lambda.min"))[-1]
sel_l <- which(beta_l != 0)
tp_l <- length(intersect(sel_l, 1:k0))
fp_l <- length(setdiff(sel_l, 1:k0))
c(selected = length(sel_l), true_pos = tp_l, false_pos = fp_l)
```

```
selected true_pos false_pos
      12       8         4
```

Python

```
from sklearn.linear_model import LogisticRegressionCV
cv_l = LogisticRegressionCV(penalty="l1", solver="saga", Cs=20, cv=10,
                             max_iter=5000).fit(X[itr], y[itr])
sel_l = np.where(cv_l.coef_[0] != 0)[0]
tp = len(set(sel_l) & set(range(k0)))
fp = len(set(sel_l) - set(range(k0)))
print({"selected": len(sel_l), "true_pos": tp, "false_pos": fp})
```

Summary

Method	Selected	True positives	False positives
Lasso (cv.glmnet)	12	8	4
COMBSS	10	9	1

In this $p \gg n$ regime, the lasso recovers the true predictors but pays a high false-positive cost — the price of ℓ_1 shrinkage in correlated designs. COMBSS returns a much sparser model that contains the truth.

Why this matters

- This is the design regime — high-dimensional, correlated predictors — where best-subset enumeration is infeasible (there are $\binom{1000}{10} \approx 2.6 \times 10^{23}$ candidate subsets).
- COMBSS finishes in seconds and returns the right support.
- The same code structure handles linear, logistic, and multinomial outcomes; only the `family` argument changes.

This is the bridge to the real biomedical applications: [Khan SRBCT](#) (multinomial, $p = 2308$) and [Rice GWAS](#) (binary, $p \approx 1.6 \times 10^5$).